

Bitcoin Market Sentiment vs Trader Performance Analysis

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 **Assignment: Data Analysis Project**

Objective

The objective of this project is to analyze how Bitcoin market sentiment (Fear, Greed, etc.) affects trader behavior and profitability. By combining trading data with the Fear & Greed Index, the goal is to identify patterns that can help in making better trading decisions.

Dataset Used

Two datasets were used in this analysis:

- **Historical Trading Data** – Includes trade details such as execution price, size, side (buy/sell), and profit/loss (PnL).
 - **Fear & Greed Index Data** – Provides daily market sentiment classification (Fear, Greed, Neutral, Extreme Greed).
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Methodology

- Cleaned and preprocessed both datasets (handled timestamps, missing values, and formats)
 - Converted timestamps into readable date format
 - Merged both datasets using the common date field
 - Performed exploratory data analysis (EDA)
 - Visualized trends using graphs to compare sentiment vs performance
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Key Findings

Profitability vs Sentiment

- Traders achieved the **highest average profit during Greed phases (~88 PnL)**
 - Profitability was moderate during **Fear (~50 PnL)**
 - Lower profits were observed during **Neutral (~22 PnL)** and **Extreme Greed (~25 PnL)**
 - This shows that normal Greed conditions are more favorable than extreme conditions
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Trading Behavior

- Trading patterns vary based on market sentiment
 - Higher activity is observed during Greed phases, indicating increased participation
 - Lower performance in Extreme Greed suggests **overconfidence and emotional trading**
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Strategic Insight

- The best trading opportunities occur during **stable Greed phases**
 - However, **Extreme Greed reduces performance**, likely due to excessive risk-taking
 - Successful traders appear to maintain consistency and avoid emotional extremes
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Conclusion

This analysis shows that market sentiment has a significant impact on trading performance. While Greed phases provide strong profit opportunities, extreme market conditions can negatively affect outcomes.

Understanding market sentiment can help traders make more informed decisions and avoid emotionally driven mistakes. A balanced and disciplined approach is more effective than reacting to extreme market behavior.